



Cloud Security Posture Management

Runbook – Remediation

Application Load Balancer HTTPS listeners should use TLS version 1.2 or higher

Control ID	ELB-041
Control Name	Application Load Balancer HTTPS listeners should use TLS version 1.2 or higher
Severity	Medium
Audience	Application Teams · Cloud Platform · Network · Cloud Security
Tooling	Wiz Cloud Security Platform
Compliance	CSA CCM v4.0.5 → IVS-07 (Network Security)
Version	1.0
Owner	Cloud Security Team
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Review Cycle	Quarterly

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1. Overview

Detection Tool	Wiz Cloud Security Platform
Wiz Rule	Cloud Configuration Rules Wiz
Detective Control	AWS Config — elb-tls-https-listeners

2. Description

This control identifies Application Load Balancer (ALB) HTTPS listeners that permit deprecated TLS versions — specifically TLS 1.0 and TLS 1.1. Both versions contain known cryptographic weaknesses and have been formally deprecated by the IETF.

Transport Layer Security (TLS) is the cryptographic protocol that secures data in transit. It provides three guarantees:

- **Confidentiality** — data is encrypted so it cannot be read in transit
- **Integrity** — data cannot be altered without detection
- **Authentication** — the server proves its identity using digital certificates

 TLS 1.0 and TLS 1.1 are deprecated and must not be used. All ALB HTTPS listeners must enforce TLS 1.2 or higher. TLS 1.3 is preferred where client compatibility allows.

3. Objective

This runbook provides standardized operational guidance for security and cloud operations teams to:

- Validate non-compliant TLS policy findings in Wiz
- Assess client compatibility and deployment risk
- Safely update ALB SSL/TLS security policies to enforce TLS 1.2 or higher
- Handle approved exceptions with documented compensating controls

4. Triage Process

 **General triage principles** — including how to assess whether a finding is valid, a false positive, or an accepted risk — are defined in the [Master Template – Remediation SOP](#). Complete that review before proceeding with the control-specific steps below.

4.1 Identify Affected Resources (Wiz)

Navigate to Wiz → Cloud Configuration Findings and filter by the rule: "Application Load Balancer HTTPS listeners should use TLS version 1.2 or higher". Export the list of affected resources.

Collect the following details for each finding:

- Load balancer name and ARN
- AWS account ID and region
- Listener ARN and port (typically 443)
- Current SSL/TLS policy name
- Application and business owner

4.2 Categorize Findings

Category	Applies to	Next Action
A – Safe to Remediate	Non-production, dev, or test load balancers with no legacy client dependencies	→ Update TLS policy immediately — no business validation required
B – Validate First	Production or externally-facing load balancers	→ Confirm client compatibility before changing. Check for legacy clients (TLS 1.0/1.1 only).
C – Exception	Known legacy integration that cannot support TLS 1.2	→ Proceed to Section 9 (Exception Process)

 Compatibility note: Enforcing TLS 1.2+ may break legacy clients that do not support modern protocols. Validate with application owners before remediating Category B listeners.

5. Manual Mitigation – AWS Console

Use this approach for one-off remediations or when the load balancer is not managed by IaC.

5.1 Update SSL/TLS Security Policy

1. Navigate to AWS Console → EC2 → Load Balancers
2. Select the affected load balancer
3. Open the Listeners and rules tab
4. Select the HTTPS listener (port 443) → click Edit listener
5. Under Security policy, select a compliant policy from the table below
6. Click Save changes
7. Repeat for all HTTPS listeners on the same load balancer
8. Verify the updated security policy appears in the Listeners tab

Recommended SSL/TLS Security Policies

Policy Name	Protocols Allowed	Status
ELBSecurityPolicy-TLS13-1-2-2021-06	TLS 1.2 and TLS 1.3	✓ Recommended — modern and secure
ELBSecurityPolicy-TLS13-1-3-2021-06	TLS 1.3 only	✓ Most secure — use if all clients support TLS 1.3
ELBSecurityPolicy-TLS-1-2-Ext-2018-06	TLS 1.2 only	✓ Acceptable — broader cipher compatibility
ELBSecurityPolicy-2016-08	TLS 1.0, 1.1, 1.2	✗ Non-compliant — do not use

✓ Changing the security policy takes effect immediately and does not drop existing connections. Only new connections are subject to the updated policy.

6. Mitigation – AWS CLI

Run the following commands in sequence. Ensure you have the necessary IAM permissions before proceeding.

Step 1 — List HTTPS listeners and current TLS policies

```
aws elbv2 describe-listeners \
  --load-balancer-arn <ALB_ARN> \
  --region <REGION> \
  --query 'Listeners[?Protocol=='HTTPS''].[ListenerArn,Port,SslPolicy]' \
  --output table
```

Step 2 — Update the listener to enforce TLS 1.2+

```
aws elbv2 modify-listener \
  --listener-arn <LISTENER_ARN> \
  --ssl-policy ELBSecurityPolicy-TLS13-1-2-2021-06 \
  --region <REGION>
```

Step 3 — Verify the change

```
aws elbv2 describe-listeners \
  --load-balancer-arn <ALB_ARN> \
  --region <REGION> \
```

```
--query 'Listeners[?Protocol==`HTTPS`].[ListenerArn,SslPolicy]' \
--output table
```

Step 4 — Test TLS enforcement (optional — requires OpenSSL)

```
# TLS 1.2 — should succeed
echo | openssl s_client -connect <ALB_DNS>:443 -tls1_2 -servername <ALB_DNS>
2>/dev/null | grep Protocol

# TLS 1.1 — should fail
echo | openssl s_client -connect <ALB_DNS>:443 -tls1_1 -servername <ALB_DNS>
2>/dev/null | grep Protocol || echo 'TLS 1.1 correctly rejected'
```

7. Mitigation Using Terraform (IaC)

Use this approach when the ALB listener is managed as Infrastructure as Code. Terraform enforces the TLS policy by setting the `ssl_policy` attribute on the `aws_lb_listener` resource.

7.1 Update SSL Policy Using Terraform

```
resource "aws_lb_listener" "https" {
  load_balancer_arn = aws_lb.app_lb.arn
  port              = 443
  protocol          = "HTTPS"

  # Enforces TLS 1.2 minimum — TLS 1.3 also accepted
  ssl_policy      = "ELBSecurityPolicy-TLS13-1-2-2021-06"
  certificate_arn = aws_acm_certificate.app_cert.arn

  default_action {
    type = "forward"
    target_group_arn = aws_lb_target_group.app_tg.arn
  }
}
```

7.2 Recommended Terraform Workflow

1. Wiz identifies the non-compliant ALB listener
2. Security team validates client compatibility and confirms Category A or B
3. Update `ssl_policy` in the Terraform listener resource

4. Run terraform plan — review the proposed changes carefully
5. Obtain peer review and approval of the plan output
6. Run terraform apply to enforce the change
7. Verify in Wiz — finding should resolve within the next scan cycle

8. Mitigation Using CloudFormation (IaC)

Use this approach when the ALB listener is managed via CloudFormation. Enforce TLS policy via the SslPolicy property on the AWS::ElasticLoadBalancingV2::Listener resource.

8.1 Update SSL Policy Using CloudFormation

```
HTTPSListener:
  Type: AWS::ElasticLoadBalancingV2::Listener
  Properties:
    LoadBalancerArn: !Ref ApplicationLoadBalancer
    Port: 443
    Protocol: HTTPS
    Certificates:
      - CertificateArn: !Ref ALBCertificate
    SslPolicy: ELBSecurityPolicy-TLS13-1-2-2021-06 # Enforces TLS 1.2+
    DefaultActions:
      - Type: forward
        TargetGroupArn: !Ref AppTargetGroup
```

8.2 Recommended CloudFormation Workflow

1. Wiz identifies the non-compliant HTTPS listener
2. Security team validates client compatibility
3. Update the SslPolicy property in the CloudFormation template
4. Create a change set and review the proposed changes carefully
5. Obtain peer review and approval of the change set
6. Execute the stack update to enforce the TLS policy

9. Document Exception

Use the exception process when a legacy integration dependency cannot be updated to TLS 1.2 within the remediation SLA (Category C from Section 4.2).

Exception Process: Submit and manage exceptions using the formal exception governance process. All required fields, approval workflows, and expiration timelines are defined there. → [Wiz CSPM Exception Governance](#)

At minimum, every exception must include:

- Business justification for continued use of deprecated TLS version
- Approver name and approval date
- Exception expiration date (maximum 180 days — must be reviewed and renewed if business need persists)
- Compensating controls in place (e.g., network-level IP allow-listing, WAF, enhanced monitoring)

10. Detective Control

Enable the AWS Config managed rule to continuously detect non-compliant ALB TLS policies independent of Wiz scan cycles.

10.1 Enable AWS Config Rule

1. Navigate to AWS Config → Rules → Add Rule
2. Search for elb-tls-https-listeners (AWS managed rule)
3. Select the rule and save

Property	Value
AWS Config Rule	elb-tls-https-listeners
Detection scope	All ALB HTTPS listeners in the account
Result	Continuous compliance visibility · independent audit evidence · reduced Wiz finding recurrence

11. Validation and Success Criteria

After completing remediation, verify success using the following checklist before closing the finding.

☐	Re-check the specific resource in Wiz and confirm the finding status is Resolved or No longer applicable.
☐	All HTTPS listeners show a compliant SSL policy (TLS 1.2+) in the AWS Console.
☐	AWS CLI describe-listeners confirms the updated ssl_policy value.
☐	OpenSSL test confirms TLS 1.2 succeeds and TLS 1.1 is rejected.

☐	Application tested — no service disruption or client errors observed.
☐	AWS Config rule elb-tls-https-listeners shows Compliant.
☐	Update the Finding Management Tool with the remediation outcome and date.
☐	If an exception was filed (Category C), confirm it is logged with an expiration date.

12. References

- [Wiz Cloud Configuration Rules](#)
- [AWS ELB Security Policies Documentation](#)
- [CSA CCM v4.0.5 Cargill Framework \(Wiz\)](#)
- [Cloud Security Navigator \(CSN\)](#)
- [Master Template – Remediation SOP](#)
- [Wiz CSPM Exception Governance](#)
- [CSPM Finding Management Tool](#)